

SUPERMARKET SALES DATA ANALYSIS

Data Cleaning, Analysis and Visualization

INTERNSHIP PROJECT REPORT

Prepared using Python, Pandas, Matplotlib, Seaborn and Google Colab.

Dataset	Supermarket Sales
Raw Records	102
Columns	11
Cleaned Records	99
Analysis Tool	Python / Google Colab

Internship Submission

1. Introduction

This project focuses on cleaning, analyzing, and visualizing supermarket sales data using Python. The raw dataset was inspected for data-quality problems such as missing values, duplicate records, inconsistent categorical values, and unusual numerical values. After cleaning, the dataset was analyzed to identify sales patterns and customer-related insights.

2. Objectives

- To understand and inspect the supermarket sales dataset.
- To identify and handle missing values.
- To identify and remove duplicate records.
- To standardize inconsistent categorical values.
- To detect and handle outliers using the IQR method.
- To analyze sales and customer patterns using Python.
- To create meaningful visualizations and summarize findings.

3. Tools and Technologies

Python: Programming language used for data analysis.

Pandas: Used for reading, cleaning and analyzing tabular data.

Matplotlib: Used to create charts.

Seaborn: Used for statistical/data visualizations.

Google Colab: Used as the development environment.

4. Methodology

The project followed an end-to-end data analytics workflow: raw data loading, dataset inspection, data cleaning, outlier detection, exploratory analysis, visualization, interpretation of results, and export of the cleaned dataset.

5. Dataset Description

The raw dataset contains **102** rows and **11** columns. The fields include invoice information, branch, city, customer type, gender, product line, unit price, quantity, total sales, payment method, and customer rating.

6. Data Cleaning

6.1 Missing Values

A total of **4** missing values were identified in the raw dataset. Missing categorical values were filled using the mode. Unit Price was filled using the median, while Rating was filled using the mean. After cleaning, the dataset contained **0** missing values.

6.2 Duplicate Records

The raw dataset contained **2** duplicate row(s). These were removed using `drop_duplicates()`. Duplicate rows after cleaning: **0**.

6.3 Inconsistent Values

The Payment column contained inconsistent capitalization, such as 'cash' and 'Cash'. The values were standardized using title-case formatting so that the same payment method would be treated consistently during analysis.

6.4 Outlier Detection

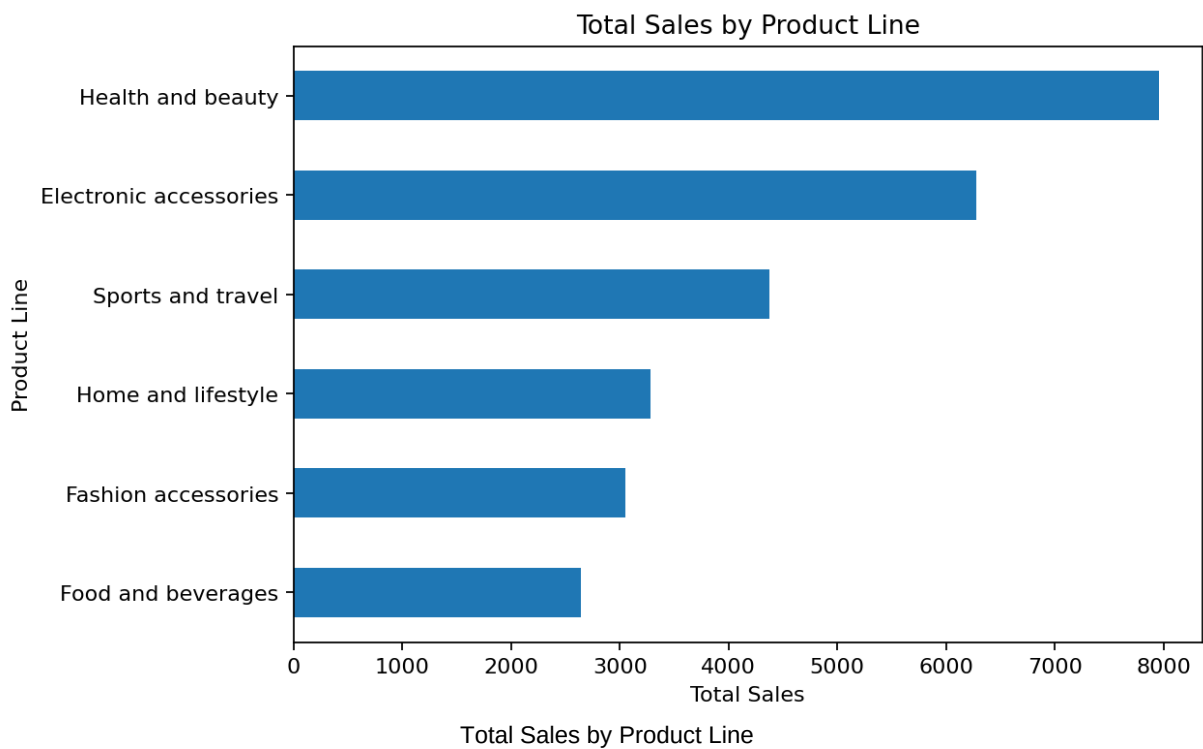
Unit Price was examined using a boxplot and the Interquartile Range (IQR) method. Q1 = 30.99, Q3 = 66.91, IQR = 35.92. The calculated acceptable range was -22.89 to 120.79. **1** outlier(s) were identified and excluded from the cleaned analysis dataset.

7. Results

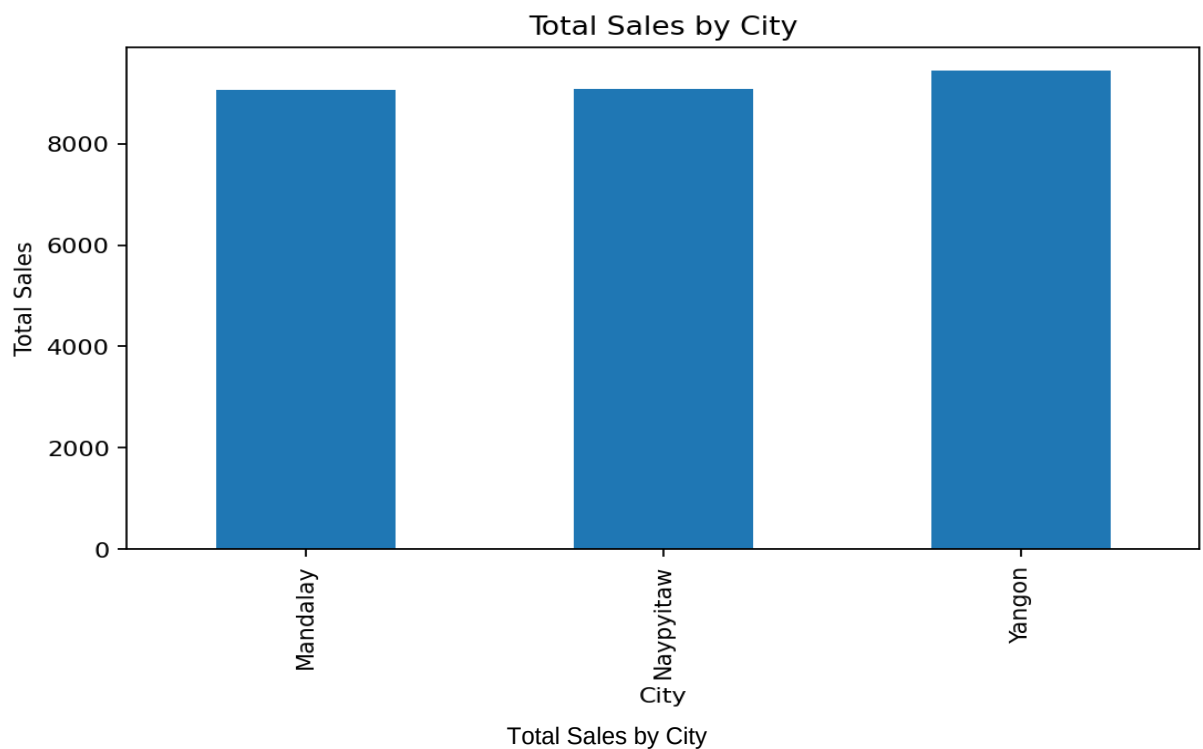
Measure	Result
Total Sales	27,588.15
Average Sales per Transaction	278.67
Total Quantity Sold	528
Highest-Selling Product Line	Health and beauty
Sales of Highest Product Line	7,959.23
Highest-Selling City	Yangon
Sales of Highest City	9,435.87
Most-Used Payment Method	Cash
Transactions Using It	37

8. Visualizations

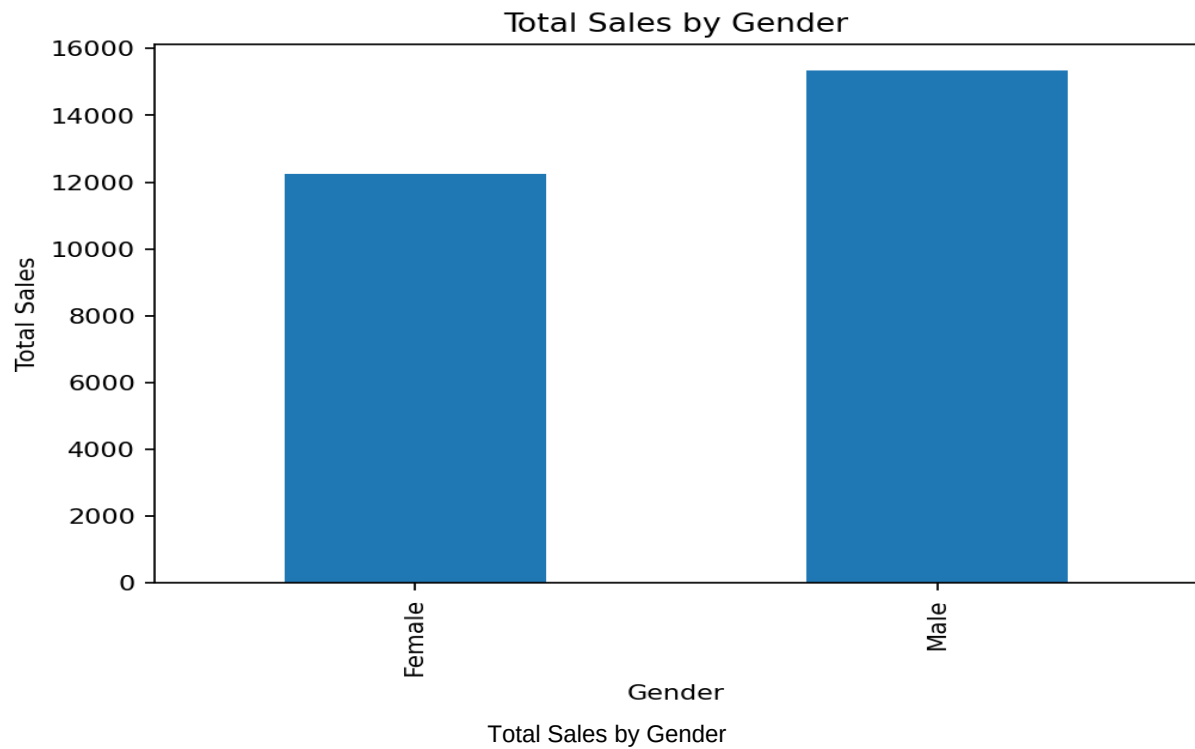
1. Total Sales by Product Line



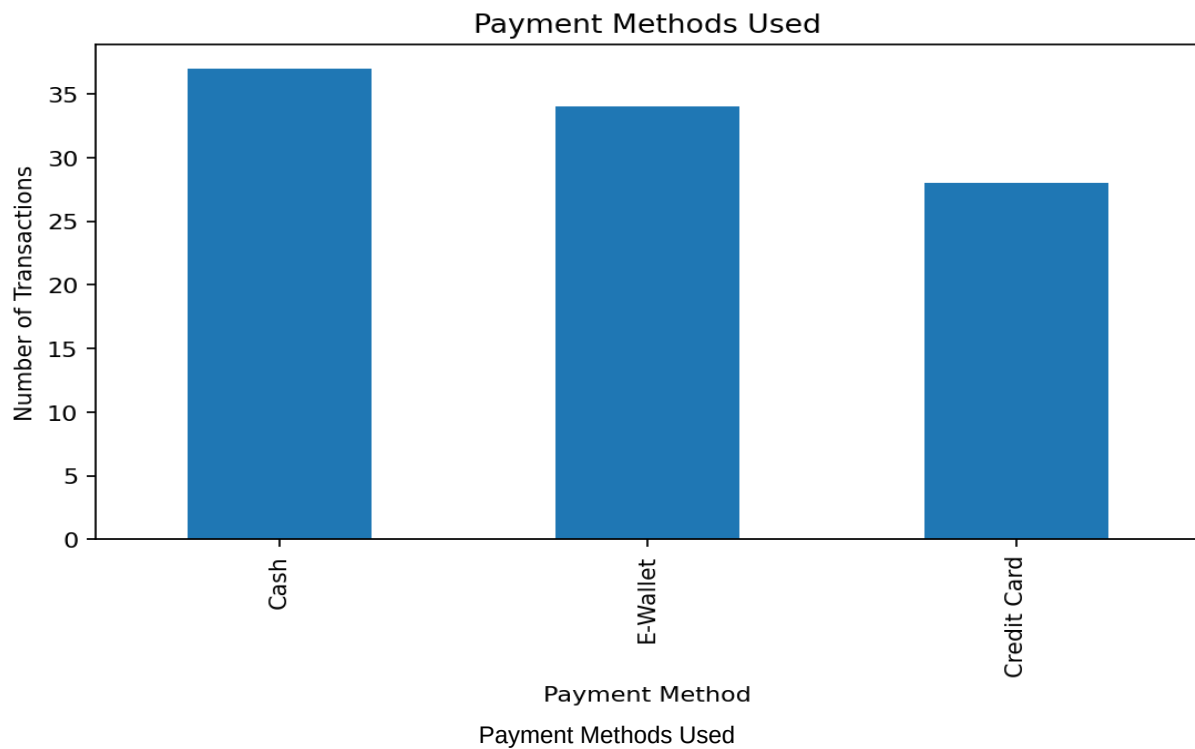
2. Total Sales by City



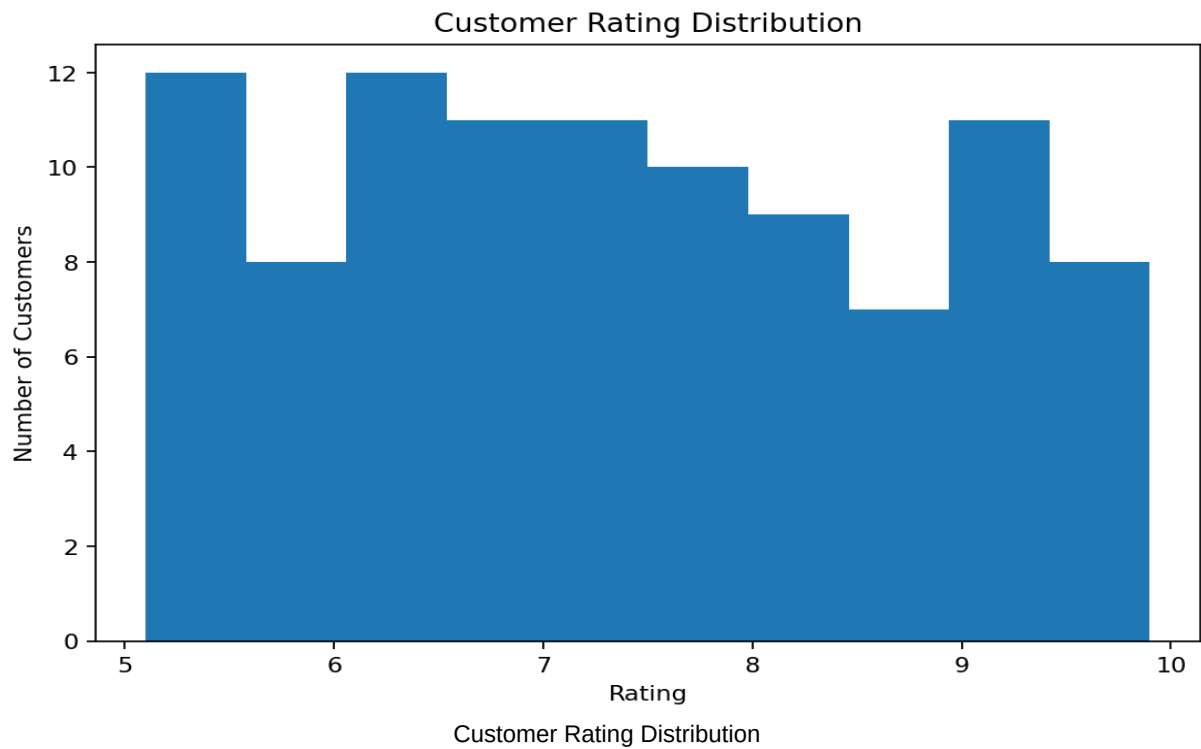
3. Total Sales by Gender



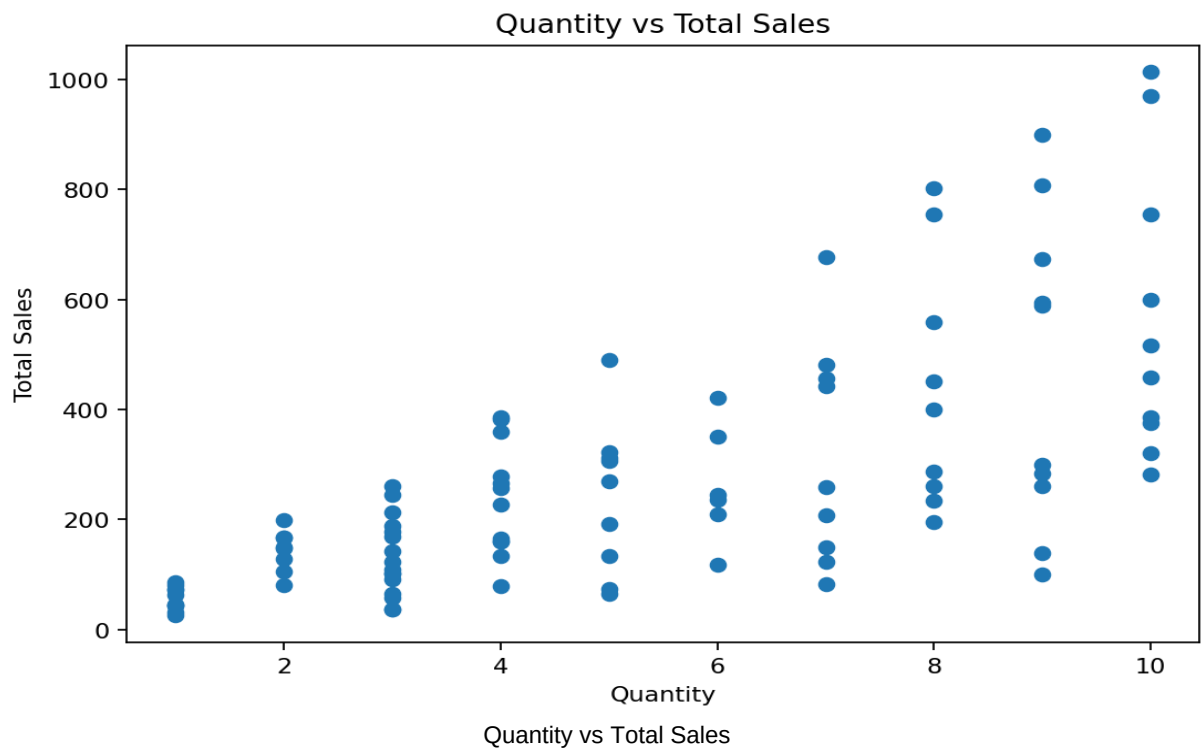
4. Payment Methods Used



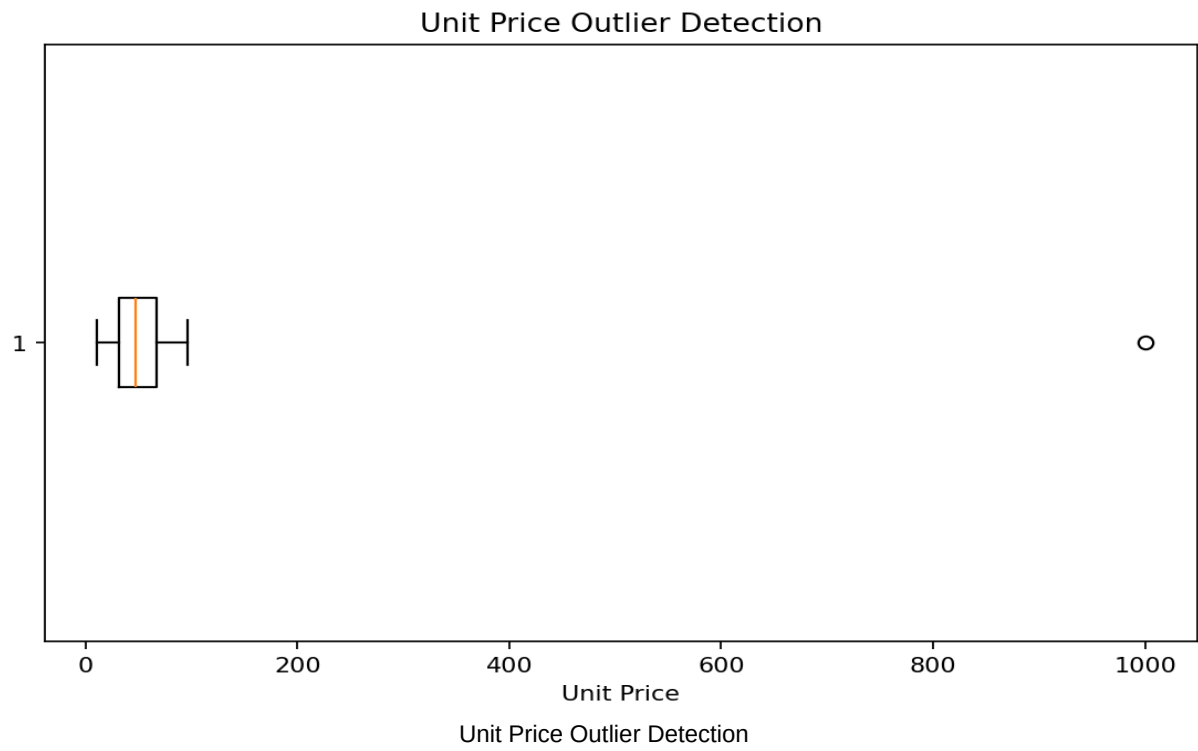
5. Customer Rating Distribution



6. Quantity vs Total Sales



7. Unit Price Outlier Detection



9. Key Insights

- **Health and beauty** was the highest-selling product line in the cleaned dataset.
- **Yangon** recorded the highest total sales among the cities analyzed.
- **Cash** was the most frequently used payment method, with 37 transactions.
- The outlier analysis showed that some Unit Price values were considerably higher than the central range and required investigation before analysis.
- Cleaning the data before analysis reduced inconsistencies and made the results more suitable for interpretation and visualization.

10. Conclusion

This internship project provided practical experience in an end-to-end data analytics workflow. The supermarket sales dataset was loaded and inspected using Pandas, followed by handling of missing values, duplicate records, inconsistent categorical values, and outliers. Sales and customer patterns were then analyzed using numerical summaries and visualizations. The final results demonstrate how Python-based data cleaning and visualization can turn raw transactional data into useful information for understanding sales performance and customer behavior.

11. Project Deliverables

The project deliverables include the original raw CSV file, the cleaned CSV file, the Google Colab notebook containing the analysis code, and this final internship report.